

A Pre-Registered Validation Framework for Geometric Perturbation Scores Across Transcriptomic, Genetic, and Morphological Profiling Data

Elliot Tower

Independent researcher
elliott@elliotttower.ai

Abstract

Background: Geometric perturbation scores—such as direction instability, which measures the angular consistency of perturbation signatures across biological contexts—rank drugs and genes by mechanism conservation across large-scale profiling datasets. These scores are biologically informative, but the raw number conflates genuine mechanism conservation with estimation artifacts: a drug scored from few cell lines may appear consistent by chance, and cross-cell-type consistency may reflect off-target programs rather than on-target mechanism. No systematic framework exists for diagnosing which confounds a given score has survived.

Results: We introduce a validation framework with seven conditions and test three empirically across 8,949 LINCS L1000 drug compounds, 1,676 Perturb-seq CRISPRi gene knockdowns, and 25,254 JUMP-CP Cell Painting morphological profiles using pre-registered hypotheses with Holm–Bonferroni correction. A variance-regularized estimator (transport-stable direction instability) outpredicted raw scores for held-out cell-line effects in all 66 valid leave-one-out folds (Wilcoxon $p = 8.2 \times 10^{-13}$; mean $\Delta\rho = +1.10$, 95% CI [1.00, 1.19]), with raw scores anti-predictive in every fold. Phenotype-projected scores correlated with on-target genetic perturbation connectivity ($\rho = 0.38$, $p < 10^{-27}$) while raw scores did not ($\rho = -0.04$). Essential-gene correction on Grassmannian subspaces revealed that universal-essential knockdowns *diverge* rather than converge across cell types ($d = 0.60$, $p_{\text{perm}} < 10^{-4}$), inverting the predicted confound and identifying proteasome/spliceosome genes as confound-dominated. Domain-specific corrections preserved global rankings in all three domains ($\rho > 0.91$) while reorganizing individual items by up to 7,155 rank positions.

Conclusions: The validation framework—define confounds, specify corrections, pre-register predictions, test whether corrections reorganize rankings or preserve them—applies to any geometric score measuring perturbation-response consistency across contexts. Corrections preserve population rankings while identifying individual perturbations most susceptible to confounding, providing a structured, pre-registerable diagnostic procedure for geometric perturbation scores.

Keywords: direction instability, perturbation scoring, validation framework, LINCS L1000, Perturb-seq, Cell Painting, variance correction, pre-registration, Fréchet variance, Grassmannian geodesic

1 Background

Geometric perturbation scores—metrics that quantify how a biological system’s response to a perturbation varies across experimental contexts—rank drugs, genes, and compounds by mechanism

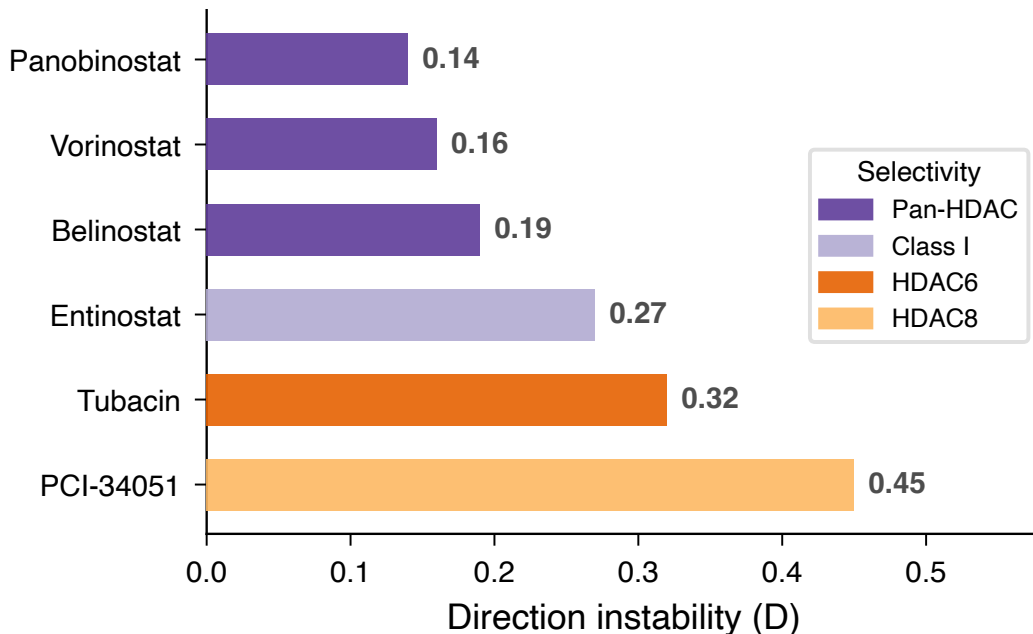


Figure 1: **HDAC selectivity gradient: calibration standard for direction instability.** Pan-HDAC inhibitors [5] (panobinostat, vorinostat, belinostat) achieve the lowest direction instability ($D = 0.14$ – 0.19); Class I-selective entinostat is intermediate ($D = 0.27$); isoform-selective HDAC6 and HDAC8 inhibitors (tubacin, PCI-34051 [6]) have the highest instability within the class ($D = 0.32$ – 0.45). This gradient, reproduced from Tower [3], provides a fine-grained rank ordering within the HDAC class: Spearman $\rho = 0.83$ between direction instability and selectivity rank across these 6 compounds after toxicity-gene correction. The HP1 “doesn’t bite” threshold ($\rho \geq 0.83$) was calibrated to this value.

conservation across large-scale profiling datasets [1, 2]. Direction instability, one such score, measures the angular consistency of perturbation signatures across cell lines: a drug whose transcriptional response points in the same direction regardless of cell type receives a low score, indicating conserved mechanism [3]. Applied to 8,949 drugs in LINCS L1000, direction instability produces biologically meaningful rankings: pan-HDAC inhibitors achieve the lowest scores ($D = 0.14$ – 0.19), with a selectivity gradient within the HDAC class (Figure 1) that tracks isoform specificity [4].

These rankings establish that the score captures genuine biological signal. They leave open a practical question: when can a practitioner trust a geometric perturbation score for downstream decisions such as hit prioritization, mechanism-of-action annotation, or cross-context prediction? A score of $D = 0.55$ means signatures point in similar directions across cell lines. It does not mean the drug acts through a specific pathway or that the observation will replicate in a new context.

Three specific confounds threaten interpretation. The raw score cannot distinguish on-target mechanism conservation from coherent off-target programs. A drug scored from few cell lines may appear consistent due to low Fréchet variance rather than genuine directional agreement. In genetic perturbation screens, universal-essential knockdowns could inflate or deflate cross-cell-type distances through a shared lethal-response subspace. No systematic procedure exists for diagnosing which of these threats a given score has survived (cf. construct-validity frameworks in measurement theory [7, 8]).

We formalize these threats through a validation framework (Figure 3): a sequence of seven con-

ditions that progressively strengthen what a geometric perturbation score supports. Each condition targets a specific confound. The ordering reflects logical dependency: a reliable measurement (Condition 2) is prerequisite for meaningful projection (Condition 5), and localized signal (Condition 4) precedes phenotype alignment (Condition 5). Satisfying additional conditions does not change the score; it narrows the set of confounds that remain viable alternative explanations.

This paper tests whether these validation conditions matter empirically across three perturbation domains: 8,949 LINCS L1000 drug compounds [2], 1,676 Perturb-seq genetic knockdowns [9], and 25,254 JUMP-CP Cell Painting morphological profiles [10]. Direction instability serves as the case study, but the procedure—define confounds, specify corrections, pre-register predictions, test whether corrections reorganize rankings or preserve them—applies to any geometric score measuring perturbation-response consistency across contexts. We report honest nulls alongside confirmations, because the pattern of where validation conditions matter—and where they do not—is itself the finding.

2 Definitions and Notation

2.1 Direction instability

For a drug tested in K cell lines, let $\mathbf{s}_1, \dots, \mathbf{s}_K \in \mathbb{R}^{978}$ denote its consensus signature vectors. Define unit-normalized signatures $\hat{\mathbf{s}}_i = \mathbf{s}_i / \|\mathbf{s}_i\|_2$. Direction instability is:

$$D = 1 - \frac{2}{K(K-1)} \sum_{i < j} \hat{\mathbf{s}}_i^\top \hat{\mathbf{s}}_j \quad (1)$$

$D = 0$ when all signatures are collinear. $D = 1$ when orthogonal on average. Cosine normalization removes the linear dependence on perturbation magnitude (Figure 2), under the assumption that signature direction and magnitude are separable—that is, that the direction a drug perturbs gene expression does not itself change with dose. Direction instability measures only direction [3]. Variance-based and estimation-based metrics conflate perturbation strength with biological signal. The separability assumption is tested empirically in the Discussion.

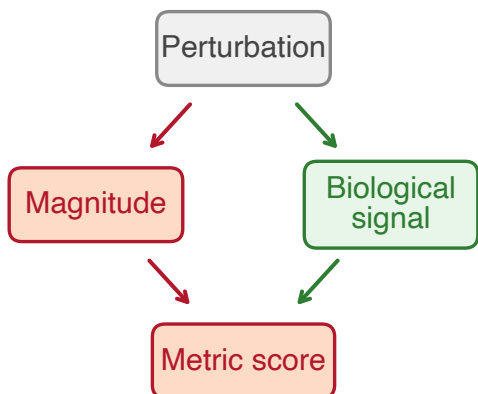
2.2 Grassmannian geodesic distance in Perturb-seq

Direction instability generalizes from vectors to subspaces. For a gene knocked down in two cell types, let $U_1, U_2 \in \text{Gr}(k, d)$ denote the top- k PCA subspaces of the perturbation response in each cell type (k = subspace dimension, d = ambient dimension). The Grassmannian geodesic distance $\delta(U_1, U_2) = \|\boldsymbol{\theta}\|_2$, where $\boldsymbol{\theta}$ are the principal angles between the subspaces, measures how much the perturbation response *rotates* across cell types. With $K = 2$ contexts, the pairwise average in Eq. 1 reduces to a single comparison, and the estimand carries no Fréchet variance or holonomy information. We therefore do not compute direction instability as defined in Eq. 1 for Perturb-seq; we compute the Grassmannian geodesic distance $\delta(U_1, U_2)$ between per-cell-type response subspaces. This is a related but distinct geometric estimand, and we label it as such throughout rather than presenting it as direction instability at $K = 2$. Perturb-seq results are interpreted as a subspace-rotation analysis, not a direct replication of the LINCS vector-cosine analysis.

2.3 Direction instability in Cell Painting morphological profiles

Cell Painting [10, 11] measures 3,180 morphological features per well across five fluorescent channels. JUMP-CP profiles 116,000 compounds across 10 imaging sources. Direction instability applies

a Confounded metrics



b Direction instability

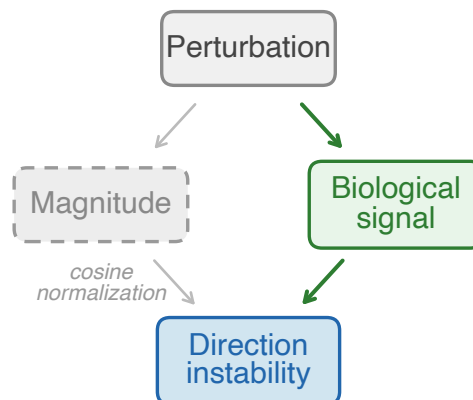


Figure 2: **Direction instability removes linear magnitude dependence.** (a) Confounded metrics (variance, subspace size) inherit effect magnitude through a shared path, producing spurious associations between perturbation strength and metric value. (b) Direction instability operates after cosine normalization, which blocks the linear magnitude path under the assumption that signature direction is independent of dose (separability). The observed association between group variable and metric reflects biological signal to the extent that this assumption holds. An empirical test of separability (dose-direction stability across 1,445 drugs) is reported in the Discussion.

directly: each compound’s morphological signature in a given source is a vector in $\mathbb{R}^{3180}$, and D measures angular consistency across sources. A subset of 743 features—cell area, compactness, eccentricity, DNA/Mito intensity, granularity, and radial distribution—reflect general cell health rather than compound-specific morphological effects. The cell-health correction removes these features before recomputing direction instability.

2.4 The validity problem

Consider two drugs with identical direction instability, $D \approx 0.56$ (mechanism-conserving regime):

- **Drug A** is a topoisomerase II inhibitor. Its conserved direction aligns with its target gene’s knockdown signature—phenotype-projected bracket is high. The consistency comes from on-target DNA-damage signaling.
- **Drug B** is a proteasome inhibitor. Its conserved direction does *not* align with its annotated target subunit’s knockdown signature—phenotype-projected bracket is near zero. The consistency comes from broad proteostasis disruption shared across cell types.

Both achieve $D \approx 0.56$. Both pass the raw metric equally. Phenotype projection (Condition 5 of the validation framework) distinguishes them: Drug A’s consistency is mechanistically on-target; Drug B’s is not. The distinction matters for downstream decisions: Drug A is a strong candidate for mechanism-based follow-up; Drug B requires additional investigation to identify what program drives its consistency before any mechanistic claim is supported. A concrete instance of this scenario (daunorubicin vs. MG-132) appears in the Discussion.

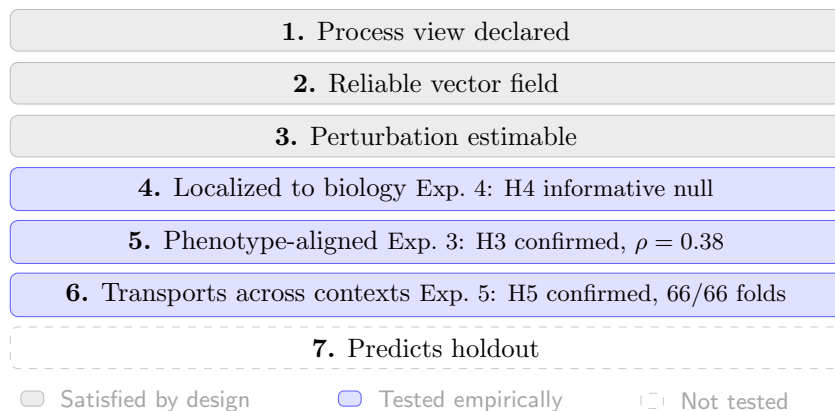


Figure 3: **The validation framework.** Seven conditions progress from measurement prerequisites (1–3) through empirically tested validation conditions (4–6) to external prediction (7). Each condition targets a specific confound. Conditions 4–6 are tested across three domains (LINCS L1000, Perturb-seq, JUMP-CP); result summaries are shown at right. Satisfying additional conditions does not change the direction-instability score—it narrows the set of confounds that remain viable.

2.5 The validation framework

The validation framework defines seven conditions for geometric perturbation scores (Figure 3), ordered by logical dependency rather than by evidence strength:

1. **Process view declared.** What constitutes a “context”?
2. **Reliable vector field.** Adequate signal-to-noise for direction to be meaningful.
3. **Perturbation estimable.** Effect isolated from background.
4. **Localized to biology.** Instability concentrates in pathway-relevant genes.
5. **Phenotype-aligned.** Conservation direction matches a known target pathway.
6. **Transports across contexts.** Score replicates in held-out contexts (low Fréchet variance).
7. **Predicts holdout.** Metric predicts an external outcome in new data.

Conditions 1–3 are satisfied by design in our datasets and are not tested further. Conditions 4–6 are tested empirically. Condition 7 (holdout prediction in a genuinely new dataset) is not tested: Experiment 5’s leave-one-cell-line-out cross-validation tests Condition 6 (transport within LINCS), not generalization to an independent dataset. A Condition 7 test would require predicting compound activity in a dataset collected independently of LINCS—such as a proprietary pharma screen or a future LINCS phase—using only validation-framework scores computed from the current data.

3 Related work

Validation frameworks for computational scores. Recent work has formalized validation procedures for computational scores: construct-validity frameworks for ML benchmarks [12, 13], the SBB principles for perturbation-prediction models [14], and the classical construct-validity tradition

in psychometrics [7, 8]. Our validation framework draws on these precedents, instantiating them for geometric perturbation scores. A mapping between the framework’s conditions and Messick’s six validity facets appears in Supplementary Table S1.

Perturbation biology. CERES [15] corrects copy-number artifacts in CRISPR screens; integrated Broad-Sanger analyses [16] document cross-study divergence. These corrections operate on dependency-score magnitudes; our contribution is orthogonal, correcting subspace geometry and showing that rank-preserving corrections carry diagnostic value. The Connectivity Map framework [1, 2] and subsequent metric evaluations [17] establish the transcriptomic-similarity scoring approach that direction instability extends. LEMUR [18] performs multivariate regression on Grassmann manifolds for multi-condition Perturb-seq; our Experiment 7 uses a simpler Grassmannian geodesic distance. Raju’s geometric coherence framework [19] applies related angular measures to single-cell CRISPR perturbations; our work differs in testing construct validity of such scores rather than using them directly for prediction. Pharmacogenomic interaction landscapes [20] and retrieval-based profile evaluation [21] provide complementary perspectives on cross-study consistency.

Metric evaluation in perturbation biology. Prior work has benchmarked connectivity and signature-similarity metrics for consistency and predictive value [2, 17, 22], validation frameworks have been proposed for ML benchmarks and computational scores [12, 13]; and recent work has shown that simple linear baselines match deep-learning perturbation predictors [23], while neural optimal transport [24] offers an alternative geometric approach to perturbation response modeling. These efforts evaluate *whether* a metric predicts a downstream label. Our contribution is orthogonal: we test *which confounds a given score has survived*—treating validation as a graded, condition-dependent diagnostic rather than a single predictive-accuracy number—and identify where those conditions affect interpretation across three perturbation modalities. We are not aware of a comparable condition-by-condition validation applied to an unsupervised geometric perturbation score with pre-registered decision criteria.

4 Methods

4.1 Data

Perturbation signatures were drawn from LINCS L1000 Phase I (GEO accession GSE92742; ~ 1.3 million profiles across ~ 50 cell lines), using Level 5 replicate-collapsed z-scores across 978 landmark genes [2, 25]. Of 77 cell lines in Phase I, approximately 50 contribute sufficient compound coverage for direction instability computation. We retained 8,949 small-molecule perturbagens tested in ≥ 5 cell lines.

Perturb-seq data derive from Replogle et al. [9]: genome-scale CRISPRi knockdowns in K562 and RPE1 cells. We extracted 1,676 genes with ≥ 30 perturbed cells in both cell types, computed per-gene expression responses over 2,073 highly variable genes, and fitted $k = 5$ PCA subspaces per gene per cell type (of these, 1,567 have matching DepMap Chronos labels for HP3; all 1,676 are used for HP1). Both screens target DepMap-essential genes; 85% of RPE1 targets overlap with K562 targets. Essentiality labels derive from DepMap Chronos scores [26]: 855 genes with Chronos < -0.5 in both K562 and RPE1 were classified as universal-essential; the remaining 712 are sub-threshold and serve as the comparison group. A label-permutation null (10,000 permutations) tested whether the observed group difference exceeds background K562/RPE1 divergence.

JUMP-CP Cell Painting profiles derive from the Cell Painting Gallery [10]: 25,254 compounds tested in ≥ 5 imaging sources after filtering (78% attrition from the 116,750 total; the majority of compounds appear in fewer than 5 sources, making direction instability—which requires multiple contexts—undefined for them). Cell-health features (743 features matching Area, Compactness, Eccentricity, DNA/Mito Intensity, Granularity, Radial Distribution, and Texture Entropy) were identified by substring matching on CellProfiler feature names; corrected direction instability was computed on the remaining 2,437 features.

Following pre-registration principles [27], all scoring functions and drug lists were committed before any real LINCS signatures were analyzed (initial freeze: commit 8d74bd0). Perturb-seq hypotheses and extended drug hypotheses (H5-full, H3-CRISPRi) were pre-registered before the correction analysis (commit f3c88ed, tag `prereg-extended-psb`, timestamped 2026-07-08T11:56:58–04:00, preceding experiment results committed at 14:42–04:00). The dose-direction separability test (Section 6.2) was pre-registered in commit 5bb6080 before execution. The repository was public at the time of all freezes.¹

Reporting guidelines. No existing reporting guideline (such as TRIPOD for clinical prediction models) directly governs a validation study of an unsupervised geometric score. We adopt TRIPOD’s spirit where applicable—pre-specified analysis plan, frozen thresholds, complete reporting of all pre-registered hypotheses including nulls—and note deviations in Table 1. We omit items (such as patient-level calibration and outcome incidence) that do not apply to a non-clinical perturbation-screen setting.

We treat the five LINCS hypotheses (H1–H5) as one test family and the three Perturb-seq hypotheses (HP1–HP3) as a second. The separation reflects genuine independence: the two domains use different experimental systems (50+ cell lines vs. two specific lines), different perturbation types (small molecules vs. CRISPRi), different geometric estimands (vector cosines in $\mathbb{R}^{978}$ vs. Grassmannian geodesics in $\text{Gr}(5, 2073)$), and share no data or parameters. A single family across all eight hypotheses would be more conservative; we consider it overly so given the absence of any shared data or analytic dependence between domains. The choice is immaterial for the confirmed results: H3 ($p < 10^{-27}$) and H5 ($p = 8.2 \times 10^{-13}$) survive Holm–Bonferroni correction even under a single 8-hypothesis family. Experiment 8 (JUMP-CP) is reported separately because it uses a third data source with a distinct geometric estimand. After Holm–Bonferroni correction within the drug family, H3 ($p < 10^{-27}$) and H5 ($p_{\text{Wilcoxon}} = 8.2 \times 10^{-13}$; also $p_{\text{perm}} < 10^{-3}$ on the preliminary 5-fold analysis) survive at $\alpha = 0.05$; H2 ($p = 0.28$) and H4 (gap = 0.018) do not; H1’s criterion-pass is an artifact. Within Perturb-seq, HP3 ($p_{\text{perm}} < 10^{-4}$) survives.

4.2 Phenotype-projected bracket (Experiment 3)

For each drug with an annotated target gene (Drug Repurposing Hub [28], 1,551 drugs), the “on-target direction” is the consensus shRNA knockdown signature of that target gene in LINCS L1000 (4,369 genes eligible, ≥ 3 hairpins each, consensus averaged across up to 20 cell lines). Phenotype-projected bracket projects the instability vector onto this on-target direction. On-target enrichment is defined as $\cos^2(\bar{\mathbf{s}}_{\text{drug}}, \bar{\mathbf{s}}_{\text{shRNA}})$: the squared cosine between each drug’s mean signature and its target gene’s shRNA consensus, measuring how much of the drug’s transcriptomic effect aligns with the genetic perturbation of its annotated target.

¹Code, pre-registration, deviation log, and all experimental scripts: <https://github.com/elliotttower/direction-instability-drug-validity>. LINCS data: GEO GSE92742; Perturb-seq: Replogle et al. [9]; JUMP-CP: Chandrasekaran et al. [10]; DepMap: Dempster et al. [26].

Table 1: Deviation log. All departures from the pre-registered plan, with rationale and timing. Full detail in `DEVIATION_LOG.md`, commit-linked.

#	Pre-registered	Actual	Why	When
1	Hypotheses frozen; gene-set method unspecified	Gene-set method declared in extended pre-reg	Original freeze omitted gene-set construction parameters for H3/H4/H5	Pre-analysis
2	H5: $\geq 4/5$ folds win	Threshold adjusts to $[0.8n]$ if < 5 valid folds	Maintains 80% win rate when rare cell lines have too few drugs; not triggered in practice (71 cell lines)	Pre-analysis
3	H3 and H4 treated as independent	Shared shRNA-consensus dependency documented	Both derive from the same genetic perturbation object; not independent tests	Pre-analysis
4	Original H5: 5 alphabetical folds	Full 66-fold leave-one-out with Wilcoxon co-primary	Extended pre-reg (commit <code>f3c88ed</code>) replaces original	Pre-analysis
5	H3-CRISPRi: three-outcome decision tree	Outcome (c) realized ($\rho = -0.12$)	Followed frozen decision tree; K562-only confound explanation pre-registered	Post-analysis

shRNA signatures have well-documented off-target effects [29], which add noise to the on-target direction and attenuate the observed correlation: the true on-target association is likely stronger than the measured $\rho = 0.38$. The shRNA consensus averages across up to 20 cell lines, producing a cell-type-agnostic on-target direction at the cost of off-target contamination. Similarly, Drug Repurposing Hub target annotations carry noise from multi-target compounds and legacy assignments; misannotation further attenuates the correlation. Polypharmacology compounds this: drugs with strong secondary targets may show phenotype projection onto the primary target even when cross-cell-line consistency is driven by a secondary mechanism, a confound that single-target annotations cannot resolve. A convergent-validity check using CRISPRi signatures (single cell type, cleaner loss-of-function but cell-type-specific) is reported in the Results.

The pre-registered prediction (H3) required Spearman $|\rho| > 0.3$ between phenotype-projected bracket and on-target enrichment, and $|\rho| < 0.15$ for raw bracket versus the same enrichment.

4.3 Variance-regularized direction instability (Experiment 5)

Transport-stable (TS) direction instability regularizes the raw score by subtracting a penalty proportional to the Fréchet variance of the signature directions across contexts:

$$TS = D - \lambda \cdot \text{Var}_{\text{Fréchet}} \quad (2)$$

where $\text{Var}_{\text{Fréchet}} = \frac{1}{K} \sum_i \arccos^2(\hat{\mathbf{s}}_i^\top \bar{\mathbf{s}})$ and $\bar{\mathbf{s}}$ is the Fréchet mean direction. The subtraction down-weights drugs whose apparent directional consistency is driven by a small number of tightly clustered contexts rather than by genuine mechanism conservation. The alternative (a ratio) would be undefined when $\text{Var}_{\text{Fréchet}} = 0$ and would amplify noise for highly consistent drugs. The penalty weight

λ was fixed at the value from Tower [3] and was not optimized for this evaluation; a sensitivity analysis across a 20-fold range of λ values is reported in Section 5.2.

We evaluate via leave-one-cell-line-out cross-validation over all cell lines with ≥ 20 eligible drugs after excluding the held-out cell line (66 valid folds out of 71 total; 5 excluded for insufficient drug count). A preliminary 5-fold analysis using alphabetically-selected cell lines (A375, A549, A673, AGS, ASC) established the rare/common regime distinction with permutation nulls; the full leave-one-out extends this to all cell lines and adds Wilcoxon signed-rank as a co-primary criterion.

The pre-registered prediction (H5) required TS to outpredict raw bracket in $\geq 80\%$ of valid folds (co-primary 1), with Wilcoxon signed-rank on per-fold $\Delta\rho$ rejecting H_0 : median ≤ 0 at $\alpha = 0.05$ (co-primary 2).

4.4 Essential-gene correction on Perturb-seq (Experiment 7)

We computed the Fréchet mean subspace of the 855 essential-gene perturbation responses (separately in K562 and RPE1) via iterative projection on $\text{Gr}(5, 2073)$ (100 iterations, convergence to $\Delta < 10^{-6}$ in Grassmannian distance; the result is insensitive to initialization because the essential-gene subspaces cluster tightly). We then projected this essential-response subspace out of each gene’s response subspace, re-orthogonalized, and recomputed geodesic distances.

Two pre-registered predictions were tested. HP1 required Spearman ρ between raw and corrected distances below 0.70 (ranking-disruptive, denoted “bites”) or ≥ 0.83 (ranking-preserving, denoted “doesn’t bite”). We use these shorthand terms throughout: a correction “bites” when it reorganizes the population ranking ($\rho < 0.70$) and “doesn’t bite” when it preserves it ($\rho \geq 0.83$). The ranking-preserving threshold of 0.83 was calibrated to the HDAC selectivity gradient (Figure 1): the finest within-class rank ordering that direction instability resolves. The ranking-disruptive threshold of 0.70 was set as a round-number boundary below which more than half the variance is unexplained ($R^2 < 0.49$), following the companion paper [3]. Values between 0.70 and 0.83 would require further analysis to interpret. HP3 predicted that universal-essential knockdowns produce higher geodesic distances than non-essential knockdowns (Cohen’s $d > 0$).

4.5 Cell-health correction on JUMP-CP (Experiment 8)

The 743 cell-health features constitute 23% of the 3,180-feature space—proportionally six times larger than the LINCS toxicity gene set (41/978, 4%) and the largest feature removal in this study. If cell-health variance dominates cross-source consistency, this correction should reorganize the ranking more substantially than the drug or genetic corrections. This makes Experiment 8 the strongest test of the thesis: if corrections don’t bite even when the removed feature set is proportionally large, the ranking is robust.

The pre-registered prediction used the same threshold: Spearman ρ between raw and corrected direction instability below 0.70 (“bites”) or ≥ 0.83 (“doesn’t bite”).

4.6 Supplementary experiments

Three additional drug-domain experiments are reported in Supplementary Experiments. **Experiment 1** (H1, toxicity correction) tests whether cytotoxic drugs masquerade as mechanism-conserving. **Experiment 2** (H2, broad-mechanism drugs) tests whether broad-target therapeutics fall below the population median. **Experiment 4** (H4, localization score) tests whether gene-set restriction improves MOA prediction. HP2 (GO pathway prediction from corrected Perturb-seq distances) is also in Supplementary Experiments.

Table 2: Pre-registered scorecard. *Criterion* = did the result meet the frozen pre-registered threshold? *Holm* = does it survive Holm–Bonferroni correction within its domain family ($\alpha = 0.05$)? *Status* = the interpretive status after both checks. The three columns earn their keep because H1 and H2 dissociate: both meet their pre-registered criteria, yet H1 is refuted (criterion-pass is an artifact) and H2 does not survive multiplicity correction. Result details for starred entries appear in Section 5.6.

	Prediction	Result	Criterion	Holm	Status
H1*	Toxicity reclassifies	$\Delta < 0.01$	Met (artifact)	—	Refuted —artifact of pre-existing high instability
H2*	Broad drugs below median	7/10 below	Met	No	Suggestive —does not survive multiplicity
H3	$ \rho_{\text{proj}} > 0.3$	$\rho = 0.376$	Met	Yes	Confirmed
H4*	AUROC gap > 0.05	Gap = 0.018	Not met	No	Informative null —domain-dependent
H5	TS wins $\geq 80\%$ folds	66/66, $p_W = 8.2 \times 10^{-13}$	Met	Yes	Confirmed
HP1	Correction bites ($\rho < 0.70$)	$\rho = 0.91$	Not met	—	Doesn’t bite
HP2*	GO prediction improves	$\Delta\rho = -0.02$	Not met	—	Doesn’t bite
HP3	Essential $d > 0$	$d = 0.60$	Met	Yes	Confirmed (inverted confound)
Exp. 8	Health correction bites	$\rho = 0.987$	Not met	—	Doesn’t bite

5 Results

5.1 Pre-registered scorecard

Table 2 summarizes all pre-registered hypotheses. Two of five drug hypotheses survive Holm–Bonferroni correction (H3, H5); H2 meets its pre-registered criterion but does not survive multiplicity correction; H1 is refuted with an artifact criterion-pass; H4 is an informative null. In the Perturb-seq domain, HP3 is confirmed; HP1 shows the correction “doesn’t bite” at the population level; HP2 is a null. Experiment 8 (JUMP-CP) shows the correction “doesn’t bite.”

5.2 Experiment 5: Variance-regularized direction instability outpredicts raw scores in all 66 held-out cell lines

Transport-stable bracket outpredicts raw bracket in all 66 valid leave-one-out folds. Both co-primary criteria are satisfied: TS wins 66/66 folds (100%, criterion $\geq 80\%$) and Wilcoxon signed-rank on per-fold $\Delta\rho$ yields $p = 8.2 \times 10^{-13}$ (criterion $p < 0.05$).

Across 66 folds, mean raw $\rho = -0.60$ (anti-predictive in every fold; range $[-0.78, -0.22]$) while mean TS $\rho = +0.50$ (positively predictive). Mean $\Delta\rho = +1.10$ (95% bootstrap CI $[1.00, 1.19]$; all 66 values positive, range $[0.17, 1.57]$). Five cell lines were excluded for having < 20 eligible drugs. Figure 4 shows the $\Delta\rho$ distribution and the frequency gradient; Table 3 shows representative folds illustrating the rare/common regime distinction.

In rarer folds (such as A673, AGS, ASC), raw bracket is strongly anti-predictive ($\rho = -0.37$ to -0.73): drugs with high raw instability in training produce held-out signatures that point *away*

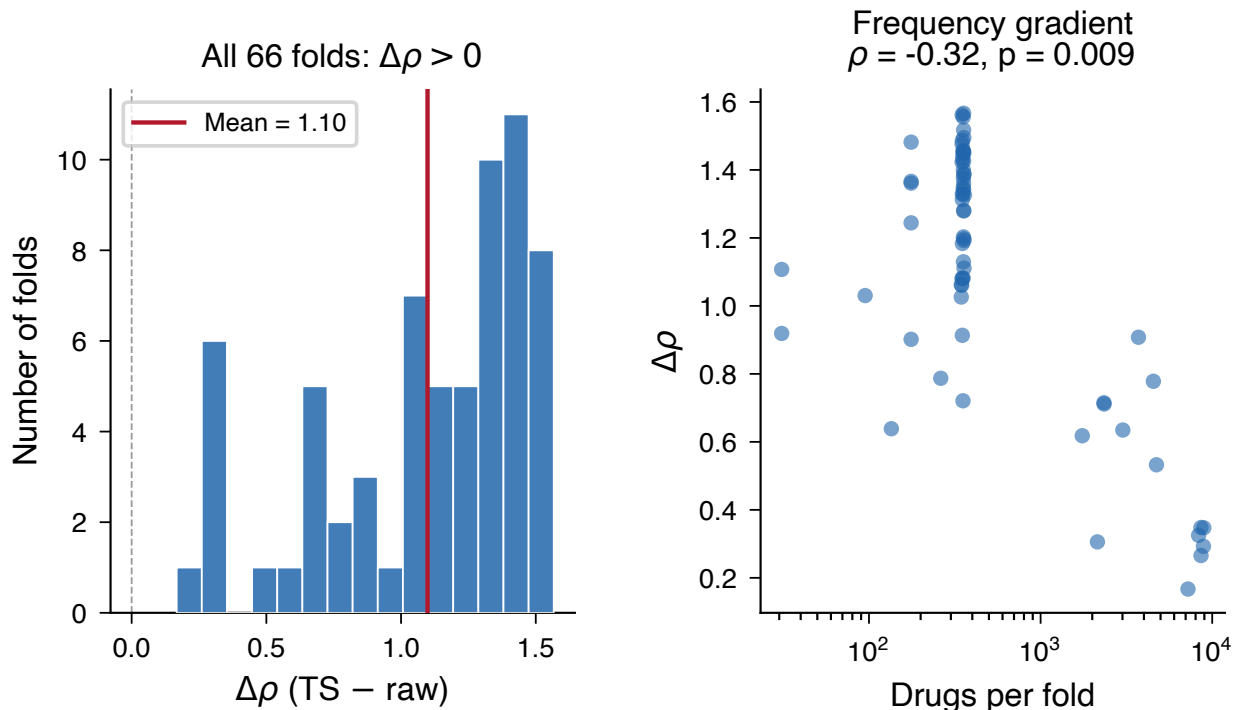


Figure 4: **Full 66-fold leave-one-out: TS advantage is universal.** **Left:** Distribution of per-fold $\Delta\rho$ (TS - raw Spearman correlation with held-out cosine). All 66 values are positive; the mean (red line) is +1.10. **Right:** Rarer cell lines (fewer drugs per fold) show larger TS advantages (Spearman $\rho = -0.32$, $p = 0.009$), confirming that the rare/common regime distinction from the preliminary 5-fold analysis generalizes across all cell lines.

from the training consensus. Transport-stable bracket reverses this, becoming positively predictive ($\rho = +0.35$ to $+0.73$). The reversal occurs because removing a single rare cell line substantially changes the consensus direction; drugs whose training instability was driven by that cell line are precisely those that TS downweights via the Fréchet penalty. This pattern holds across all 66 folds: the frequency-improvement correlation (Spearman $\rho = -0.32$, $p = 0.009$) confirms that rarer cell lines show larger TS advantages.

In common folds (8,000+ drugs), both metrics remain anti-predictive. TS improves over raw ($\rho = -0.155$ vs. -0.48 for A375; -0.201 vs. -0.49 for A549), but both correlations stay negative. TS “wins” in these folds by losing less badly, not by becoming positively predictive. The consensus direction is stable when computed from many cell lines, leaving less room for the Fréchet penalty to improve prediction.

Raw direction instability is anti-predictive because it conflates two sources of angular consistency: genuine mechanism conservation (signal) and the geometric tightness of the consensus direction around a small number of contexts (noise). Drugs with low raw DI in training tend to have tight consensus directions that are sensitive to removing any single cell line—precisely the drugs that fail to predict held-out effects. This is a failure of the raw estimator when applied to leave-one-out prediction without variance correction, not a failure of direction instability as a concept.

Before conducting the full 66-fold leave-one-out, we tested whether the TS advantage could be algebraic using a variance-preserving permutation null on the preliminary 5-fold analysis. Held-out

Table 3: Leave-one-cell-line-out cross-validation (selected folds from the full 66-fold analysis). Transport-stable (TS) bracket outpredicts raw bracket in all folds. In common folds (8,000+ drugs), both metrics are anti-predictive but TS is less so. In rarer folds, TS reverses the sign entirely. Full 66-fold results: mean $\Delta\rho = +1.10$, Wilcoxon $p = 8.2 \times 10^{-13}$.

Held-out cell	N_{drugs}	ρ_{raw}	ρ_{TS}	Winner
A375	8,328	-0.480	-0.155	TS
A549	8,888	-0.494	-0.201	TS
A673	357	-0.725	+0.725	TS
AGS	356	-0.712	+0.714	TS
ASC	2,345	-0.365	+0.346	TS

signatures are shuffled across drugs while training data (raw DI, Fréchet variance, TS, consensus vector) remain fixed, and held-out cosines are recomputed against each drug’s original consensus. Under this null (1,000 permutations), TS wins only 0.1 of 5 folds on average with mean $\Delta\rho = -0.12$ (raw DI *outpredicts* TS when drug-specific signal is broken). The observed advantage falls entirely outside the null distribution ($p_{\text{perm}} < 10^{-3}$). The full 66-fold Wilcoxon test ($p = 8.2 \times 10^{-13}$) provides independent confirmation that the advantage is drug-specific.

Lambda sensitivity. The penalty weight $\lambda = 1.0$ was inherited from the companion paper and not optimized for this evaluation. To test whether the H5 result depends on this specific value, we repeated the full 66-fold cross-validation at $\lambda \in \{0, 0.25, 0.5, 0.75, 1.0, 1.5, 2.0, 3.0, 5.0\}$. At $\lambda = 0$, TS reduces to raw bracket exactly; the sanity check confirms 0/66 wins (strict inequality on identical vectors is deterministically false). At $\lambda = 0.25$, the smallest non-zero value, TS already wins 54/66 folds (82%, Wilcoxon $p = 9.5 \times 10^{-8}$). All eight non-zero λ values pass the pre-registered criterion ($\geq 80\%$ wins and Wilcoxon $p < 0.05$), far exceeding the “ ≥ 5 of 8” success threshold. For $\lambda \geq 0.75$, TS wins all 66 folds ($p = 8.2 \times 10^{-13}$). The 66/66 win rate saturates at a ceiling; the per-fold $\Delta\rho$ distribution provides finer resolution of the advantage magnitude across λ values. Figure 5 summarizes the sensitivity analysis. The H5 result is robust to penalty weight choice across a 20-fold range.

5.3 Experiment 3: Phenotype-projected bracket separates on-target from off-target instability

Among 795 drugs with annotated targets and matched shRNA consensus signatures, phenotype-projected bracket correlates with on-target enrichment at Spearman $\rho = 0.376$ ($p = 4.8 \times 10^{-28}$). Raw bracket shows no correlation ($\rho = -0.043$, $p = 0.22$). Both pre-registered criteria are satisfied.

H3 confirms Condition 5 of the validation framework. Raw direction instability measures whether a drug’s signatures point in consistent directions, but this consistency could arise from any coherent program. Projecting onto the genetic perturbation direction of the annotated target isolates the mechanistically relevant component. The near-zero raw correlation shows that global instability carries no on-target information; the strong projected correlation shows that the on-target projection recovers it.

The gene-set construction avoids researcher degrees of freedom: the shRNA consensus is an empirical object from the same LINCS platform, requiring no pathway database selection. H3 and H4 share this shRNA dependency (documented in DEVIATION_LOG.md entry 3) and are

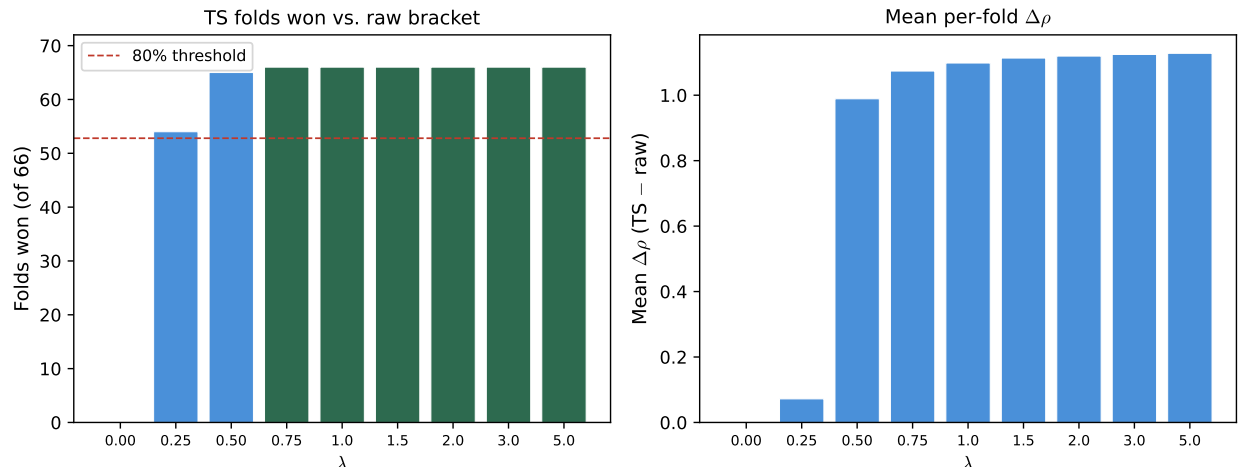


Figure 5: **Lambda sensitivity analysis for transport-stable bracket.** **Left:** Number of folds (out of 66) where TS outpredicts raw bracket, by λ . The dashed line marks the pre-registered 80% threshold (52.8 folds). At $\lambda = 0$ (sanity check), $\text{TS} \equiv \text{raw}$ and wins 0 folds. All eight non-zero λ values pass the criterion. **Right:** Mean per-fold $\Delta\rho$ ($\text{TS} - \text{raw}$) increases steeply from $\lambda = 0$ to $\lambda = 0.5$ and plateaus above $\lambda = 1.0$, indicating diminishing returns from stronger variance penalization.

not independent tests. The phenotype-projected bracket is applicable only to drugs with target annotations—1,551 of 8,949 (17%). For the remaining 82% of drugs with unknown targets, the validation framework provides other conditions (transport stability, localization) that do not require target annotations.

As a pre-registered convergent-validity check, we recomputed phenotype-projected bracket using CRISPRi loss-of-function signatures [9] as the on-target direction. The projected-bracket correlation with on-target enrichment did not reach significance (Spearman $\rho = -0.12$, $p = 0.16$, $n = 131$), corresponding to pre-registered outcome (c). On the 114-drug matched subset (drugs with both ground truths), shRNA projection yields $\rho = 0.42$ while CRISPRi projection yields $\rho = -0.14$, confirming that the ground-truth source—not the drug set—drives the difference.

The pre-registered secondary criterion points to the likely cause: raw bracket correlated positively with CRISPRi enrichment ($\rho = +0.27$, $p = 0.002$) while projected bracket did not—the reverse of the shRNA pattern. This reversal is consistent with the pre-registered K562-only confound: CRISPRi signatures derive from a single hematopoietic cell type, so the CRISPRi “on-target direction” captures cell-type-specific effects rather than cross-context mechanism—a confound that multi-cell-line shRNA consensus avoids by averaging across ~ 20 cell lines. One plausible explanation for the positive raw-bracket/CRISPRi correlation, which we did not test directly, is that K562 expresses many drug targets at high levels, producing spurious alignment between raw instability and K562-specific signatures.

CRISPRi does not provide independent convergent validity for H3. The shRNA-based result stands on its original evidence rather than being reinforced. An alternative concern is that the H3 correlation is partly tautological: both drug signatures and shRNA ground-truth signatures derive from the same LINCS platform and largely overlapping cell lines, so shared platform-specific variance could inflate the association. The near-zero raw-bracket correlation ($\rho = -0.04$) argues against this: if platform sharing drove the result, raw bracket should also correlate with shRNA enrichment, since it uses the same signatures. The correlation emerges only after phenotype pro-

jection, suggesting it reflects on-target alignment rather than shared platform noise.

HDAC-removal sensitivity. Pan-HDAC inhibitors are the calibration standard for direction instability (Figure 1), so the H3 correlation could be driven by this single drug class. To test this, we removed all 20 HDAC-targeting drugs (spanning HDAC1, HDAC2, and HDAC6 targets; full list in Supplementary Experiments) and recomputed the phenotype-projected correlation. The result is essentially unchanged: $\rho = 0.3759$ ($n = 775$, $p = 2.0 \times 10^{-27}$), compared with $\rho = 0.3756$ on the full 795-drug set ($\Delta\rho = 3 \times 10^{-4}$). A synonym check for “histone deacetylase,” “class I HDAC,” “pan-HDAC,” and “sirtuin” confirmed no drugs were missed by the primary filter.

Removing the 20 drugs with lowest raw bracket (most directionally consistent regardless of class) also preserves the correlation ($\rho = 0.387$, $p = 4.8 \times 10^{-29}$). Removing the 20 drugs with highest projected bracket—a direct test of whether a handful of phenotype-aligned outliers carry the signal—yields $\rho = 0.350$ ($p = 9.8 \times 10^{-24}$). All three removal analyses exceed the pre-registered threshold of $\rho > 0.20$. Bootstrap 95% CI on the full set is $[0.315, 0.433]$, excluding zero. The H3 correlation is broad-based, not driven by HDAC inhibitors or a small number of extreme drugs.

5.4 Experiment 7: Essential-gene correction preserves ranking but reveals confound-dominated core-machinery genes

Universal-essential knockdowns have *higher* Grassmannian geodesic distances than non-essential knockdowns (mean 2.80 vs. 2.67, Cohen’s $d = +0.60$, Mann–Whitney $p < 10^{-33}$; $n = 1,567$ genes with both geodesic distances and DepMap labels, of which 855 universal-essential and 712 sub-threshold). Cohen’s $d = 0.60$ is a medium effect—the distributions overlap substantially, and individual genes span the full range in both groups. The finding is that the *group means* differ in the opposite direction from the predicted confound, not that essentiality status determines geodesic distance.

Essential genes do not produce a shared response across cell types; they *diverge*—the same gene’s lethal disruption manifests through different transcriptomic programs in K562 and RPE1 (Figure 6). This is consistent with cell-line-specific essentiality effects documented in CERES [15] and integrated Broad–Sanger analyses [16]: the functional consequences of losing a core-machinery gene depend on the cell’s lineage and metabolic context.

A label-permutation null (shuffling essential/non-essential labels across the 1,567 genes, 10,000 permutations) confirms that this divergence exceeds background K562/RPE1 divergence: the observed difference (0.123) falls outside the null 95% interval $[-0.021, 0.022]$, $p_{\text{perm}} < 10^{-4}$. The null preserves the geodesic-distance distribution while breaking only the label assignment, so the $\sim 6\times$ exceedance of the null half-width rules out the library-overlap confound (85% target overlap between screens).

Spearman ρ between raw and corrected geodesic distances is 0.91 ($p < 10^{-100}$). The global ranking is preserved.

The correction’s value is at the item level. The top 100 rank-shifting genes are enriched for spliceosomal complex assembly (Fisher’s exact OR = 8.9, $p = 1.3 \times 10^{-7}$) and proteasomal catabolic processes (OR = 4.1, $p = 6.7 \times 10^{-5}$). Proteasome subunits (*PSMA3*, *PSMB6*, *PSMC4*) shift 750–900 rank positions; spliceosome components (*SF3B1*, *SNRPG*, *DDX23*) shift 660–820 positions. The group-mean comparison (essential vs. non-essential, Cohen’s $d = 0.60$) provides supporting context: the distributions overlap substantially, and essentiality status is a weak per-gene classifier. HP3 is best understood as a shift in group-level geometry localized to identifiable core-machinery complexes, not a gene-level discriminator. These complexes are the genes whose cross-cell-type

geometry is most dominated by the shared essential-response axis; the correction provides a concrete action: flag them before interpreting their transport scores.

5.5 Experiment 8: Cell-health correction has negligible effect in JUMP-CP

Among 25,254 compounds, corrected direction instability tracks raw at Spearman $\rho = 0.987$ ($p < 10^{-100}$). By the pre-registered criterion, this “doesn’t bite”—the strongest result for ranking robustness across all three domains.

The 743 health features constitute 23% of the 3,180-feature space—proportionally six times larger than the drug toxicity gene set (41/978, 4%)—yet produce *less* ranking impact ($\rho = 0.987$ vs. $\rho > 0.99$ for LINCS). That a proportionally larger feature removal barely moves the ranking is stronger evidence for the thesis than the drug case: cell-health features carry less of the cross-source variance in morphological signatures than stress genes carry in transcriptomic signatures. As noted in Section 4.1, the 78% attrition from the full JUMP-CP dataset (116,750 to 25,254 compounds) means the retained set is biased toward well-replicated compounds; the result may not generalize to sparsely-profiled compounds.

One explanation for the negligible effect: in morphological space, “health” and “mechanism” features are less separable than in transcriptomic space. A drug that alters cell shape changes both health-associated features (area, compactness) and mechanism-specific features (texture, radial distribution) simultaneously, because morphological effects are inherently coupled. In 978-gene transcriptomic space, stress-response genes and pathway-specific genes are more separable, which is why the LINCS toxicity correction has a larger (though still small) effect.

The near-perfect global correlation coexists with individual compounds shifting up to 7,155 rank positions, identifying those whose apparent morphological consistency is health-feature-dominated. The cell-health feature definition (substring matching on CellProfiler names) is a limitation: a partition based on correlation with viability assays would be more principled and could yield different correction magnitudes.

5.6 Summary of supplementary experiments

Full methods, tables, and discussion appear in Supplementary Experiments.

H1 (toxicity correction). Among 13 cytotoxic drugs, most are already high-instability ($D > 0.87$ for 9 of 13); toxicity correction shifts values by < 0.01 . The predicted failure mode—cytotoxicity masquerading as mechanism conservation—does not hold. Cytotoxicity produces genuinely context-dependent signatures in 978-gene space.

H2 (broad-mechanism drugs). Seven of 10 broad-mechanism drugs fall below the population median. Pan-HDAC inhibitors dominate ($D = 0.49$ – 0.55). The pre-registered second prediction—that kinase inhibitors would fall above the median—failed. The sorting axis is target breadth, not clinical specificity.

H4 (localization). Macro-averaged AUROC gap is $+0.018$ (below $+0.05$ threshold). Localization improves prediction for intracellular enzyme inhibitors (topoisomerase: $+0.49$, CDK: $+0.26$) but hurts for receptor-mediated drugs (acetylcholine: -0.20 , PPAR: -0.19). Condition 4 is domain-dependent.

HP2 (pathway function prediction). GO Jaccard overlap has near-zero correlation with geodesic distance ($|\rho| < 0.025$) for both raw and corrected forms. Geometric similarity of perturbation responses is distinct from functional pathway relatedness.

6 Discussion

6.1 Validation conditions preserve rankings while diagnosing confounds

The central empirical finding: validation conditions do not reorganize the direction-instability ranking across any of the three domains tested. This thesis was genuinely at risk—had any correction produced $\rho < 0.70$ while simultaneously improving prediction of an external outcome, the raw ranking would have been wrong. That no correction crossed this threshold across three domains ($\rho > 0.91$ in all cases) is the main result. The Perturb-seq result involves a degenerate $K = 2$ pairwise comparison rather than genuine $K \geq 5$ angular consistency; the qualitative conclusion holds across domains, but the genetic domain tests a weaker geometric estimand.

The value of the validation framework is at the item level. A rank-preserving correction (“doesn’t bite”) admits two interpretations: the targeted confound is absent, or the confound is present but rank-preserving by construction. These are distinguishable when item-level rank shifts are examined: proteasome/spliceosome genes shift 660–900 positions in Perturb-seq while the global correlation holds at $\rho = 0.91$. The population-level ρ and the item-level rank shifts together provide richer diagnostics than either alone.

The three headline results each address a different confound. Variance-regularized direction instability (H5) outpredicts raw scores in all 66 leave-one-out folds ($p = 8.2 \times 10^{-13}$), with raw scores anti-predictive in every fold—the strongest single result. Phenotype projection (H3) decomposes direction instability into on-target and off-target components ($\rho = 0.38$ vs. -0.04); a CRISPRi convergent-validity check does not independently confirm this finding, likely because K562-only CRISPRi signatures are confounded by cell-type-specific effects. Essential-gene correction identifies proteasome/spliceosome genes as confound-dominated while the population ranking holds.

In practice, the framework changes which items proceed to follow-up. Failing a condition does not downgrade a drug’s ranking; it flags the drug for further investigation before a mechanistic claim is supported. Consider two scenarios:

Drug repurposing triage. Daunorubicin (topoisomerase II inhibitor, 19 cell lines) and MG-132 (proteasome inhibitor, 16 cell lines) have nearly identical raw direction instability ($D = 0.557$ vs. 0.565 , $\Delta = 0.008$). Raw bracket ranks them equivalently—both in the mechanism-conserving regime. Phenotype-projected bracket separates them: daunorubicin scores 11.1 with on-target enrichment 0.24 (its consistency aligns with *TOP2A* knockdown); MG-132 scores 3.5 with enrichment < 0.001 (its consistency does not project onto *PSMB1* knockdown). One interpretation, consistent with proteasome-inhibitor pharmacology, is that MG-132 produces consistent transcriptomic effects through broad proteostasis disruption rather than through the specific program of its annotated target subunit. A practitioner using raw bracket alone would treat both drugs identically. After Condition 5 validation, daunorubicin is prioritized for mechanism-based follow-up; MG-132 requires investigation of what program drives its consistency. This illustrative pair was selected by matching raw bracket ($|\Delta D| < 0.05$) and ranking by projected-bracket divergence; individual drug-level scores carry uncertainty from cell-line sampling ($n = 16$ – 19).

Perturb-seq hit triage. Among 100 genes with low geodesic distance, 15 shift >500 rank positions after essential-gene correction (proteasome, spliceosome components). These are flagged as potentially confound-dominated. The remaining 85 genes have the same transport scores but stronger validation evidence—they are prioritized for functional validation. The confidence upgrade changes which genes proceed to follow-up.

6.2 What direction instability measures

Under the DAG of Figure 2, cosine normalization blocks the linear magnitude path from perturbation strength to the metric. Direction instability operates in a specific regime: it compares signatures *across cell lines at a fixed dose*. In this regime, magnitude variation arises from cell-line-specific pharmacodynamics (how strongly each cell line responds), and cosine normalization removes it. The separability assumption—that signature direction is independent of response magnitude—holds trivially when the dose is fixed, because the “magnitude” that varies across cell lines is response amplitude, not concentration.

A stronger version of separability would require direction to be independent of dose concentration. A pre-registered empirical test (commit 5bb6080) assessed this stronger claim across 6,169 drug×cell-line pairs profiled at ≥ 2 doses (1,445 drugs, 52,920 dose-pair comparisons). The median within-pair cosine between dose-level signatures was 0.19, and 86.5% of pairs fell below 0.5—both pre-registered criteria failed (thresholds: median > 0.7 , $< 20\%$ below 0.5).

These low cosines are dominated by signature-level noise rather than genuine dose-dependent direction changes. Of dose-level groups in this analysis, 80% contain a single replicate; in 978 dimensions, two single-replicate z-score vectors have expected cosine near zero even when the underlying true direction is identical. A diagnostic control confirms this: near-identical doses (ratio < 1.01 , such as 9.99 vs. 10.00 μM —the same intended dose with measurement imprecision) yield median cosine = 0.11, establishing the noise floor. The overall median of 0.19 is barely above it. The pre-registered criteria cannot be met at this replicate depth, and the test lacks statistical power to assess separability.

The separability assumption therefore remains empirically unresolved: the test was underpowered for single-replicate data in 978 dimensions. A properly powered test would require ≥ 3 replicates per dose level and a within-dose replicate baseline—conditions that the LINCS Level 5 archive does not provide for most drug×cell×dose combinations.

Two additional boundary conditions affect the DAG’s blocking claim. Near-zero-norm signatures amplify noise under unit-normalization (guarded in code at $\|s\| < 10^{-10}$). The Fréchet variance—the basis for the transport-stable correction (Eq. 2)—is part of the DAG’s structure rather than a component of D itself. We have not proven identification with do-calculus; the DAG is a heuristic justification for cosine normalization as a confound-blocking adjustment, not a formal identification theorem.

6.3 Cross-domain consistency

The same geometric object—angular deviation of perturbation-response directions across contexts—captures mechanism conservation in three unrelated domains (Table 4). All three exhibit the same pattern: the relevant validity condition preserves the global ranking while identifying individual perturbations whose scores are most confound-dominated.

The domain-specific failure modes differ: in drugs, cytotoxicity produces high-instability signatures; in genetics, universal-essential knockdowns produce high-distance subspaces (divergent lethal programs); in morphology, cell-health features contribute a small fraction of cross-source variance. All three invert the naive prediction that “violence mimics transport.”

6.4 Generalizability of the validation procedure

The validation framework was tested on one geometric perturbation score (direction instability) across three biological domains. The *procedure*—define interpretive threats, specify corrections that target each threat, pre-register predictions about whether corrections reorganize rankings,

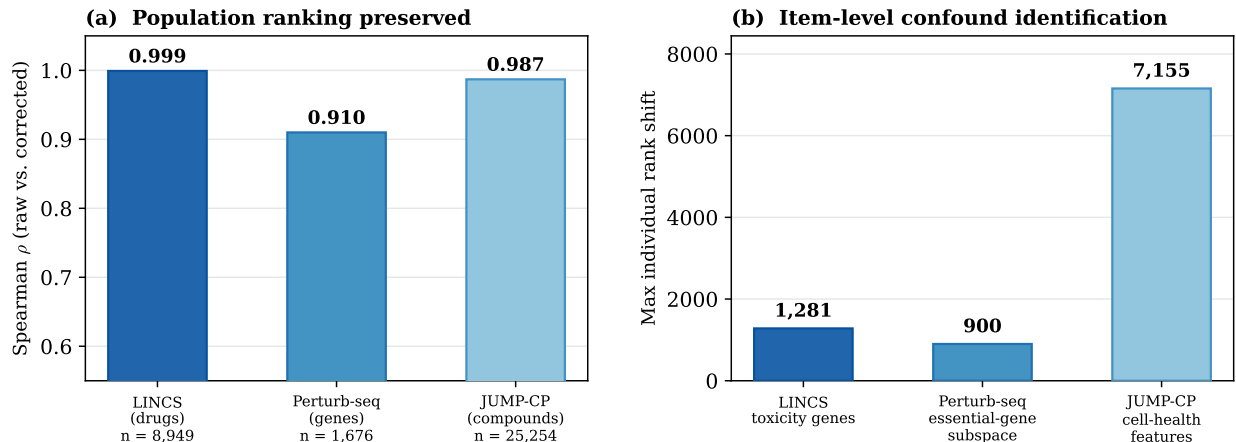


Figure 6: **Cross-domain empirical summary.** (a) Spearman ρ between raw and corrected scores across three domains. All three exceed $\rho = 0.90$: corrections preserve population rankings. Sample sizes shown below each bar. (b) Maximum individual rank shift per domain, showing that rank-preserving corrections still identify specific confound-dominated items. JUMP-CP produces the largest individual shifts (7,155 positions) from a proportionally large feature removal (23% of features), yet population $\rho = 0.987$.

test empirically—applies to other geometric scores that measure perturbation-response consistency, such as connectivity scores [2], morphological similarity indices, and Grassmannian distance metrics. What would change are the domain-specific conditions: a connectivity score would require different confound corrections than direction instability, but the same structure of threat identification, correction, and pre-registered testing applies. The effect size for phenotype projection ($\rho = 0.38$, explaining $\sim 14\%$ of variance) is moderate. For comparison, connectivity-score benchmarks in LINCS typically report Spearman correlations of $\rho = 0.2$ – 0.4 between metric and biological outcome [17], placing the phenotype-projection result at the upper end of the range achievable with unsupervised geometric scores on transcriptomic data.

6.5 Scope and limitations

Direction instability measures angular consistency of perturbation signatures across contexts. It does not measure therapeutic efficacy, in-vivo translatability, or clinical specificity. The validation framework is a validation framework for perturbation-score interpretation, not a surrogate endpoint. The consequential validity facet (whether confidence upgrades change practitioner decisions) is illustrated via the daunorubicin/MG-132 case study but is not empirically tested; a user study or decision-simulation experiment would be needed to validate this facet.

The drug experiments H3–H5 depend on shRNA knockdown signatures as ground-truth on-target directions. shRNA has well-documented off-target effects, which add noise to the ground truth. A CRISPRi convergent-validity check did not independently confirm the phenotype-projection result ($\rho = -0.12$, $p = 0.16$, $n = 131$); the matched 114-drug comparison (shRNA $\rho = 0.42$ vs. CRISPRi $\rho = -0.14$) points to the pre-registered K562-only confound (CRISPRi signatures from a single cell type are poor on-target directions for drugs acting in non-hematopoietic contexts). The shRNA H3 result stands on its original evidence without independent reinforcement. Drugs with unannotated targets (7,398 of 8,949) are excluded from H3 and H4, limiting these conditions to 17% of the dataset. LINCS measures 978 landmark genes, restricting the toxicity gene set to the

Table 4: Cross-domain summary. All three domains exceed the “doesn’t bite” threshold ($\rho \geq 0.83$). The LINCS HDAC gradient ρ measures fine-grained ordering within a single drug class (6 compounds in the selectivity gradient, Figure 1); the global 8,949-drug $\rho > 0.99$. H5 (transport stability) tests predictive validity rather than rank preservation; the 66-fold leave-one-out result (Table 2) provides the strongest domain-specific evidence for LINCS.

Domain	Correction	N	ρ
LINCS (HDAC gradient)	Toxicity genes	6	0.83
LINCS (global)	Toxicity genes	8,949	>0.99
Perturb-seq	Essential-gene subspace	1,676	0.91
JUMP-CP	Cell-health features	25,254	0.99

landmark panel.

The Perturb-seq analysis uses $K = 2$ cell types, providing no Fréchet variance, holonomy, or transport stability test. Both screens target DepMap-essential genes, so the HP3 “non-essential” group consists of sub-threshold essentials rather than genuinely non-essential genes. A genome-wide screen with genuine non-essential controls would be sharper, though the label-permutation null rules out the library-overlap confound.

The transport-stable bracket formula (Eq. 2) is a heuristic linear combination of direction instability and Fréchet variance, not a principled geometric operation on the signature manifold. The penalty weight $\lambda = 1.0$ was inherited from the companion paper [3] and was not optimized for this evaluation. A pre-registered sensitivity analysis confirmed that all eight non-zero λ values tested (0.25–5.0) pass the H5 criterion, so the result is not specific to $\lambda = 1.0$ (see §5.2). A ratio or a geodesic-based combination might perform differently. The simplicity of the linear combination is intentional: it admits straightforward interpretation of the penalty weight and enables the lambda sensitivity analysis reported in §5.2.

A pre-registered dose-direction stability test (6,169 drug $\times$ cell-line pairs) produced low within-pair cosines (median 0.19; 86.5% below 0.5), failing both pre-registered criteria. Post-hoc analysis revealed this failure is dominated by signature-level noise: 80% of dose-level groups contain a single replicate, and near-identical doses (ratio < 1.01) yield median cosine = 0.11, establishing a noise floor close to the overall result. The test lacks statistical power to distinguish genuine dose-dependent direction changes from single-replicate noise in 978 dimensions. No a priori power analysis preceded the pre-registration; replicate depth should have been assessed before committing to specific thresholds. Direction instability operates at a fixed dose across cell lines, a regime where the relevant separability condition—that cell-line response magnitude does not affect direction—is weaker than dose-independence and holds trivially. The stronger claim (direction independent of dose) remains empirically unresolved.

LINCS L1000 profiles have known batch effects across plates and production sites. Batch-driven systematic direction differences would inflate direction instability for drugs profiled across batches, producing a confound not addressed by any validity condition tested here.

Deviation entry 1 (Table 1) documents that the original pre-registration freeze omitted gene-set construction parameters; these were specified in the extended pre-registration. The gap between freezes created a potential researcher degree of freedom in gene-set construction, partially mitigated by the empirical construction (shRNA consensus from the same platform requires no pathway-database selection).

The Perturb-seq subspace dimension ($k = 5$) was fixed; geodesic distances depend on this choice, and a principled selection criterion (such as an explained-variance threshold) would strengthen the analysis.

The validation framework draws an analogy to Bradford Hill criteria for causal inference [30]. These criteria are individually neither necessary nor sufficient for establishing causation; they are heuristic guidelines. The validation framework shares this character: it is a structured set of proof obligations rather than a formal deductive system. Each condition eliminates a specific threat, but satisfying all tested conditions does not *guarantee* that a mechanistic claim is correct—it guarantees that specific alternatives have been ruled out.

JUMP-CP cell-health features were defined by substring matching rather than by correlation with an independent viability assay. The 78% attrition from the full dataset biases the retained set toward well-replicated compounds.

7 Conclusions

Variance-regularized direction instability outpredicts raw scores in all 66 leave-one-out folds ($p = 8.2 \times 10^{-13}$, mean $\Delta\rho = +1.10$), with raw scores anti-predictive in every fold. Phenotype projection separates on-target from off-target consistency ($\rho = 0.38$ vs. -0.04). Essential-gene correction identifies proteasome/spliceosome genes as confound-dominated while preserving the population ranking ($\rho = 0.91$). These results hold across three perturbation domains—8,949 LINCS drugs, 1,676 Perturb-seq knockdowns, 25,254 JUMP-CP compounds—with domain-specific failure modes that invert naive predictions in each case.

The validation framework tested here—define confounds, specify corrections, pre-register predictions, test whether corrections reorganize rankings or preserve them—applies to any geometric score measuring perturbation-response consistency across contexts. Localization helps for intracellular enzyme targets but not receptor-mediated drugs, constraining where specific conditions apply. The domain-specific corrections differ, but the pattern is the same: correction preserves population rankings while identifying individual perturbations most susceptible to confounding. The framework provides a structured, pre-registerable procedure for diagnosing which confounds a geometric perturbation score has survived.

List of abbreviations

AUROC: area under the receiver operating characteristic curve; CRISPRi: CRISPR interference; DI: direction instability; GO: Gene Ontology; HDAC: histone deacetylase; JUMP-CP: Joint Undertaking for Morphological Profiling – Cell Painting; LINCS: Library of Integrated Network-Based Cellular Signatures; LOO: leave-one-out; MOA: mechanism of action; PCA: principal component analysis; shRNA: short hairpin RNA; TS: transport-stable (variance-regularized) direction instability.

Declarations

Ethics approval and consent to participate

Not applicable. This study uses only publicly available datasets (LINCS L1000, Perturb-seq, JUMP-CP, DepMap) with no human subjects research.

Consent for publication

Not applicable.

Availability of data and materials

All data used in this study are publicly available: LINCS L1000 (GEO GSE92742), Perturb-seq [9], JUMP-CP [10], DepMap [26]. Code, pre-registration commits (SHA 8d74bd0 for initial freeze; SHA f3c88ed, tag prereg-extended-psb, for extended pre-registration; SHA 5bb6080 for dose-direction separability test), deviation log, and all experimental scripts are available at <https://github.com/elliotttower/direction-instability-drug-validity>. The repository was public before the pre-registration freeze. A permanent archive is deposited at Zenodo (DOI: <https://doi.org/10.5281/zenodo.21284088>).

Competing interests

The author declares no competing interests.

Funding

This work received no external funding.

Authors' contributions

ET conceived the study, designed the validation framework, conducted all experiments, and wrote the manuscript.

Use of AI-assisted technologies

Claude (Anthropic) was used as a writing and analysis assistant during manuscript preparation. The author directed all analytical decisions and verified all numerical results against source data. The validation framework, pre-registered hypotheses, experimental results, and interpretations are the author's own. The author takes full responsibility for the content. No AI tool is listed as an author.

Acknowledgements

Not applicable.

Supplementary information

Additional file 1: Supplementary Experiments and Extended Results. Full methods, tables, and discussion for Experiments 1, 2, and 4 (H1 toxicity correction, H2 broad-mechanism drugs, H4 localization score) and HP2 (GO pathway prediction from corrected Perturb-seq distances). Includes the complete HDAC drug list used in the sensitivity analysis and Supplementary Table S1 (Messick validity-facet mapping).

References

- [1] Justin Lamb, Emily D Crawford, David Peck, et al. The Connectivity Map: using gene-expression signatures to connect small molecules, genes, and disease. *Science*, 313(5795): 1929–1935, 2006. doi: 10.1126/science.1132939.
- [2] Aravind Subramanian, Rajiv Narayan, Steven M Corsello, David D Peck, Ted E Natoli, Xiaodong Lu, Joshua Gould, John F Davis, Andrew A Tubelli, Jacob K Asiedu, et al. A next generation connectivity map: L1000 platform and the first 1,000,000 profiles. *Cell*, 171(6): 1437–1452, 2017.
- [3] Elliot Tower. Direction instability predicts cross-cell-line drug mechanism transport in LINCS L1000. *Zenodo*, 2026. doi: 10.5281/zenodo.21213700.
- [4] Elliot Tower. A direction-instability atlas across drugs, genes, and morphology. *Zenodo*, 2026. Preprint.
- [5] Jessica E Bolden, Melissa J Peart, and Ricky W Johnstone. Anticancer activities of histone deacetylase inhibitors. *Nature Reviews Drug Discovery*, 5(9):769–784, 2006. doi: 10.1038/nrd2133.
- [6] Sriram Balasubramanian, Joseph Ramos, Weibo Luo, et al. A novel histone deacetylase 8 (HDAC8)-specific inhibitor PCI-34051 induces apoptosis in T-cell lymphomas. *Leukemia*, 22(5):1026–1034, 2008. doi: 10.1038/leu.2008.9.
- [7] Lee J Cronbach and Paul E Meehl. Construct validity in psychological tests. *Psychological Bulletin*, 52(4):281–302, 1955. doi: 10.1037/h0040957.
- [8] Samuel Messick. Validity of psychological assessment: validation of inferences from persons’ responses and performances as scientific inquiry into score meaning. *American Psychologist*, 50(9):741–749, 1995. doi: 10.1037/0003-066X.50.9.741.
- [9] Joseph M Replogle, Reuben A Saunders, Angela N Pogson, Jeffrey A Hussmann, Alexander Lenail, Alina Guna, Lauren Mascibroda, Eric J Wagner, Karen Adelman, Gila Lithwick-Yanai, et al. Mapping information-rich genotype-phenotype landscapes with genome-scale Perturb-seq. *Cell*, 185(14):2559–2575, 2022.
- [10] Srinivas Niranj Chandrasekaran, Jeanelle Ackerman, Eric Alix, D Michael Ando, John Arevalo, Melissa Berber, Nicolas Bornholdt, Alexander A Boulgakov, et al. JUMP Cell Painting dataset: morphological impact of 136,000 chemical and genetic perturbations. *bioRxiv*, pages 2023–03, 2023.
- [11] Mark-Anthony Bray, Shantanu Singh, Han Han, Chadwick T Davis, Blake Borgeson, Cathy Hartland, Maria Kost-Alimova, Sigrun M Gustafsdottir, Christopher C Gibson, and Anne E Carpenter. Cell painting, a high-content image-based assay for morphological profiling using multiplexed fluorescent dyes. *Nature Protocols*, 11(9):1757–1774, 2016. doi: 10.1038/nprot.2016.105.
- [12] Timo Freiesleben and Sebastian Zezulka. The benchmarking epistemology: Construct validity for evaluating machine learning models. *arXiv preprint arXiv:2510.23191*, 2025.
- [13] Noah Watson et al. Measuring what matters: Construct validity in LLM benchmarks. *arXiv preprint arXiv:2511.04703*, 2025.

- [14] SBB Consortium. Principles for evaluating virtual cell perturbation models. *bioRxiv*, 2026. doi: 10.64898/2026.04.20.719650.
- [15] Robin M Meyers, Jordan G Bryan, James M McFarland, Barbara A Weir, Ann E Sizemore, Han Xu, Neekesh V Dharia, Phillip G Montgomery, Glenn S Cowley, Sasha Pantel, et al. Computational correction of copy number effect improves specificity of CRISPR–Cas9 essentiality screens in cancer cells. *Nature Genetics*, 49(12):1779–1784, 2017. doi: 10.1038/ng.3984.
- [16] Clare Pacini, Joshua M Dempster, Isabella Boyle, Emanuel Gonçalves, Hanna Najgebauer, Emre Karakoc, Dieudonne van der Meer, Lukas Poolber, Alice Sber, Aviad Tsherniak, et al. Integrated cross-study datasets of genetic dependencies in cancer. *Nature Communications*, 12:1661, 2021. doi: 10.1038/s41467-021-21898-7.
- [17] Qiao Li, Trang Pham, Ming Li, et al. A comprehensive evaluation of connectivity methods for L1000 data. *Briefings in Bioinformatics*, 21(6):2194–2205, 2020. doi: 10.1093/bib/bbz129.
- [18] Constantin Ahlmann-Eltze and Wolfgang Huber. Analysis of multi-condition single-cell data with latent embedding multivariate regression. *Nature Genetics*, 57:1–10, 2025. doi: 10.1038/s41588-024-01996-0.
- [19] Prashant C. Raju. Geometric coherence of single-cell CRISPR perturbations reveals regulatory architecture and predicts cellular stress. *arXiv preprint arXiv:2604.16642*, 2026.
- [20] Francesco Iorio, Theo A Knijnenburg, Daniel J Vis, Graham R Bignell, Michael P Menden, Michael Schubert, Nanne Aben, Emanuel Gonçalves, Syd Barthorpe, Howard Lightfoot, et al. A landscape of pharmacogenomic interactions in cancer. *Cell*, 166(3):740–754, 2016. doi: 10.1016/j.cell.2016.06.017.
- [21] Alexandr A Kalinin, John Arevalo, Loan Vulliard, Hillary Tsang, Nicolas Bornholdt, Srinivas Niranj Chandrasekaran, et al. A versatile information retrieval framework for evaluating profile strength and similarity. *Nature Communications*, 16:5181, 2025. doi: 10.1038/s41467-025-60306-2.
- [22] Francesco Napolitano, Diego Carrella, Barbara Mandriani, Sandra Pisonero-Vaquero, Francesco Sirci, Angelo Facchiano, and Diego di Bernardo. Reconciling multiple connectivity scores for drug repurposing. *Briefings in Bioinformatics*, 22(6):bbab161, 2021. doi: 10.1093/bib/bbab161.
- [23] Constantin Ahlmann-Eltze, Wolfgang Huber, and Simon Anders. Deep-learning-based gene perturbation effect prediction does not yet outperform simple linear baselines. *Nature Methods*, 22(8):1657–1661, 2025. doi: 10.1038/s41592-025-02772-6.
- [24] Charlotte Bunne, Stefan G Stark, Gabriele Gut, Jacobo Sarabia del Castillo, Kjong-Van Lehmann, Lucas Pelkmans, Andreas Krause, and Gunnar Ratsch. Learning single-cell perturbation responses using neural optimal transport. *Nature Methods*, 20(11):1759–1768, 2023. doi: 10.1038/s41592-023-01969-x.
- [25] Alexandra B Keenan, Sherry L Jenkins, Kathleen M Jagodnik, Simon Koplev, Edward He, Denis Torre, Zichen Wang, Anders B Dohlman, Moshe C Silverstein, Alexander Lachmann, et al. The library of integrated network-based cellular signatures NIH program: System-level cataloging of human cells response to perturbations. *Cell Systems*, 6(1):13–24, 2018. doi: 10.1016/j.cels.2017.11.001.

- 757 [26] Joshua M Dempster, Isabella Boyle, Francisca Vazquez, David E Root, Jesse S Boehm,
758 William C Hahn, Aviad Tsherniak, and James M McFarland. Chronos: a cell population dy-
759 namics model of CRISPR experiments that improves inference of gene fitness effects. *Genome*
760 *Biology*, 22(1):343, 2021. doi: 10.1186/s13059-021-02540-7.
- 761 [27] Brian A Nosek, Charles R Ebersole, Alexander C DeHaven, and David T Mellor. The pre-
762 registration revolution. *Proceedings of the National Academy of Sciences*, 115(11):2600–2605,
763 2018. doi: 10.1073/pnas.1708274114.
- 764 [28] Steven M Corsello, Joshua A Bittker, Zihan Liu, Joshua Gould, Patrick McCarren, Jodi E
765 Hirschman, Stephen E Johnston, Anita Vrcic, Bang Wong, Mariam Khan, et al. The drug
766 repurposing hub: A next-generation drug library and information resource. *Nature Medicine*,
767 23(4):405–408, 2017.
- 768 [29] Ian Smith, Peyton G Greenside, Ted Natoli, David L Lahr, David Wadden, Itay Tirosh,
769 Rajiv Narayan, David E Root, Todd R Golub, Aravind Subramanian, and John G Doench.
770 Evaluation of RNAi and CRISPR technologies by large-scale gene expression profiling in the
771 Connectivity Map. *PLOS Biology*, 15(11):e2003213, 2017. doi: 10.1371/journal.pbio.2003213.
- 772 [30] Austin Bradford Hill. The environment and disease: Association or causation? *Proceedings of*
773 *the Royal Society of Medicine*, 58(5):295–300, 1965. doi: 10.1177/003591576505800503.